

COMBINED HELICAL GEAR + SHAFT STRESS ANALYSIS (ALL GEARS)

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clc; clear; close all;
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INPUT PARAMETERS

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% Power / Loading
Power_hp = 10;           % [hp]
PLV = 1000;              % [ft/min]

% Pitch Diameter
GTI = [50, 42, 38, 36, 35, 32];
GTO = [20, 28, 32, 34, 35, 38];
module = 3.5;
PDI = GTI * module;
PDO = GTO * module;
dp = PDI.*GTI;

% Shaft
T_Nm = 500;
T_Nmm = T_Nm * 1000;
L_shaft = 350;           % [mm]
d_shaft_o = 40;           % [mm]
d_shaft_i = 34;           % [mm]

% Convert AGMA tangential load to SI
Wt_lb = (33000 * Power_hp) / PLV; % [lb]
Wt = (2*T_Nmm)./dp;           % [N]

% Gear Geometry
psi_deg = 15;
phi_n_deg = 14.5;
psi = deg2rad(psi_deg);
phi_n = deg2rad(phi_n_deg);

% Gear Locations
x = [17.5, 102.5, 132.5, 217.5, 247.5, 332.5]; % [mm]

% Diameters [mm]
d_p = [115.9509, 126.8213, 130.4448, 137.6917, 152.1856, 181.1733];

% Teeth
Np_all = [50, 42, 38, 36, 35, 32];
Ng_all = [20, 28, 32, 34, 35, 38];
GearRatios = Ng_all./Np_all;
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% AGMA Factors
HB = 250;
K_O = 1.5;
K_v = 1.3;
K_s = 1.0;
K_B = 1.0;

C_mc = 1; C_pf = 0.1; C_pm = 1; C_ma = 0.15; C_e = 1;
K_m = 1 + C_mc*(C_pf*C_pm + C_ma*C_e);

F = 25 / 25.4; % face width [in]

% Material Strength
S_t_MPa = 0.533 * HB + 88.3;
S_t = S_t_MPa * 145.038; % psi
Y_N = 1.3558 * (1e7)^(-0.0178);

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Loop through Bending and Gear Stress Calculations for each Gear set

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% -----

for i = 1:length(d_p)

    fprintf('\n===== \n');
    fprintf('                GEAR %d ANALYSIS \n', i);
    fprintf('===== \n');

    %% ----- GEAR FORCES -----
    phi_t = atan(tan(phi_n) / cos(psi));
    rad2deg(phi_t);
    Wt_i = Wt(i);
    Wa = Wt_i * tan(psi);
    Wr = Wt_i * tan(phi_t);

    %% ----- SHAFT ANALYSIS -----

    % Reactions (Horizontal)
    R_Bx = (Wt_i * x(i)) / L_shaft;
    R_Ax = Wt_i - R_Bx;

    Mx = R_Ax * x(i);

    % Vertical plane
    M_axial = Wa * (d_p(i)/2);

    R_By = (Wr * x(i) + M_axial) / L_shaft;
    R_Ay = Wr - R_By;

    My = abs(R_Ay * x(i));

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% Total bending
M_total = sqrt(Mx.^2 + My.^2);

% Geometry (hollow shaft)

I = (pi/64) * (d_shaft_o^4 - d_shaft_i^4);
J = (pi/32) * (d_shaft_o^4 - d_shaft_i^4);
c = d_shaft_o / 2;

Area = (pi/4) * (d_shaft_o^2 - d_shaft_i^2);

% Stresses
sigma_bending = (M_total * c) / I;
tau_torsion    = (T_Nmm * c) / J;
sigma_axial    = Wa / Area;

sigma_x = sigma_bending + sigma_axial;

% Von Mises stress
sigma_vm = sqrt(sigma_x.^2 + 3 * tau_torsion.^2);

%% ----- AGMA GEAR STRESS -----

Np = Np_all(i);
dp_in = d_p(i) / 25.4;

Pd = Np / dp_in;

Nv = Np / (cos(psi)^3);
J_spur = 0.484 - (2.87 / Nv);
J = J_spur / cos(psi);

sigma_t = (Wt_lb * K_O * K_v * K_s * K_m * K_B * Pd) / (F * J);

S_F = (S_t * Y_N) / sigma_t;

%% ----- OUTPUT -----

fprintf('--- Gear Forces ---\n');
fprintf('Wt: %.2f N | Wa: %.2f N | Wr: %.2f N\n', Wt_i, Wa, Wr);

fprintf('\n--- Shaft Results ---\n');
fprintf('Mx: %.2f Nmm | My: %.2f Nmm\n', Mx, My);
fprintf('Bending Stress: %.2f MPa\n', sigma_bending);
fprintf('Axial Stress: %.2f MPa\n', sigma_axial);
fprintf('Torsion Stress: %.2f MPa\n', tau_torsion);
fprintf('Von Mises: %.2f MPa\n', sigma_vm);

fprintf('\n--- AGMA Gear Results ---\n');
fprintf('Bending Stress: %.2f psi\n', sigma_t);

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fprintf('Allowable: %.2f psi\n', S_t * Y_N);
fprintf('Safety Factor: %.3f\n', S_F);
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end
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                        GEAR 1 ANALYSIS
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--- Gear Forces ---
Wt: 114.29 N | Wa: 30.62 N | Wr: 30.60 N
--- Shaft Results ---
Mx: 1900.00 Nmm | My: 419.94 Nmm
Bending Stress: 0.65 MPa
Axial Stress: 0.09 MPa
Torsion Stress: 83.24 MPa
Von Mises: 144.18 MPa
--- AGMA Gear Results ---
Bending Stress: 20001.87 psi
Allowable: 32700.11 psi
Safety Factor: 1.635
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                        GEAR 2 ANALYSIS
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--- Gear Forces ---
Wt: 161.97 N | Wa: 43.40 N | Wr: 43.37 N
--- Shaft Results ---
Mx: 11739.90 Nmm | My: 2337.31 Nmm
Bending Stress: 3.99 MPa
Axial Stress: 0.12 MPa
Torsion Stress: 83.24 MPa
Von Mises: 144.24 MPa
--- AGMA Gear Results ---
Bending Stress: 15719.76 psi
Allowable: 32700.11 psi
Safety Factor: 2.080
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                        GEAR 3 ANALYSIS
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--- Gear Forces ---
Wt: 197.86 N | Wa: 53.02 N | Wr: 52.98 N
--- Shaft Results ---
Mx: 16291.90 Nmm | My: 3052.94 Nmm
Bending Stress: 5.52 MPa
Axial Stress: 0.15 MPa
Torsion Stress: 83.24 MPa
Von Mises: 144.29 MPa
--- AGMA Gear Results ---
Bending Stress: 14043.07 psi
Allowable: 32700.11 psi
Safety Factor: 2.329
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                        GEAR 4 ANALYSIS
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--- Gear Forces ---
Wt: 220.46 N | Wa: 59.07 N | Wr: 59.03 N
--- Shaft Results ---
Mx: 18152.40 Nmm | My: 2332.88 Nmm
Bending Stress: 6.09 MPa
Axial Stress: 0.17 MPa
Torsion Stress: 83.24 MPa
Von Mises: 144.31 MPa
--- AGMA Gear Results ---
Bending Stress: 12719.39 psi
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Allowable: 32700.11 psi

Safety Factor: 2.571

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GEAR 5 ANALYSIS

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--- Gear Forces ---

Wt: 233.24 N | Wa: 62.50 N | Wr: 62.45 N

--- Shaft Results ---

Mx: 16905.46 Nmm | My: 1163.49 Nmm

Bending Stress: 5.64 MPa

Axial Stress: 0.18 MPa

Torsion Stress: 83.24 MPa

Von Mises: 144.30 MPa

--- AGMA Gear Results ---

Bending Stress: 11244.36 psi

Allowable: 32700.11 psi

Safety Factor: 2.908

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GEAR 6 ANALYSIS

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--- Gear Forces ---

Wt: 279.02 N | Wa: 74.76 N | Wr: 74.70 N

--- Shaft Results ---

Mx: 4638.67 Nmm | My: 5191.91 Nmm

Bending Stress: 2.32 MPa

Axial Stress: 0.21 MPa

Torsion Stress: 83.24 MPa

Von Mises: 144.20 MPa

--- AGMA Gear Results ---

Bending Stress: 8784.06 psi

Allowable: 32700.11 psi

Safety Factor: 3.723